Proceedings of ASME Turbo Expo 2026
Turbomachinery Technical Conference and Exposition
GT2026
June 15-19, 2026, Milan, Italy

GT2026-176316

Attendee Read-Only Copy

TOWARDS GENERALISABLE PREDICTIVE HEALTH MODELS FOR DIVERSE TURBOFAN ENGINES VIA KNOWLEDGE TRANSFER

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ABSTRACT

Predictive models for aircraft engines are being developed to forecast its health condition utilizing available data. In-service engine data will be the most valuable information to build such a model but may not always be accessible or sufficient. A pre-trained model can be a possible alternative, if it can be extended to new and other engine families. The pre-trained model needs to be transferable and versatile for this objective, so that the knowledge acquired by the model can be transferred. The objective of this paper is to demonstrate knowledge transfer by evaluating a predictive engine health model across different applications of engines. A predictive engine health model was developed using Machine Learning techniques applied to Engine Health Monitoring (EHM) data gathered from the fleets of three different turbofan engines. Each model was built exclusively by training on the EHM data from each fleet of turbofan engines. The pre-trained models made predictions for their source (baseline case), and for other fleets of turbofan engines (transfer knowledge cases). The results show that the transfer knowledge cases have mean Root-Mean-Square-Error values, between 5°C and 8°C, which is about 2 times higher than the baseline case. The baseline case, represented by smaller interquartile range (IQR), between 0.4°C and 1.6°C, have less variation in the prediction results than that of the transfer knowledge cases which have IQR values ranging from 1.6°C to 6°C. Even so, the results confirmed that the models are versatile and can be transferred across turbofan engines. Moreover, a generative predictive engine health framework can be developed with the potential to scale across diverse gas turbine engine classes and applications.

Keywords: Turbofan Engine, Engine Health Monitoring, Transfer Learning, Machine Learning

NOMENCLATURE

b	bias vector
c	cell state
f	forget gate state
h	hidden layer state
i	input gate state
o	output gate state
W	weight matrix
RMSE	root mean-squared error
tanh	hyperbolic tangent activation function
σ	sigmoid activation function

1. INTRODUCTION

Predictive models could estimate the health condition of aircraft engines as it leads to optimized maintenance and reduction of unscheduled removals. In recent years, advances in Artificial Intelligence (AI) and Machine Learning (ML) have encouraged the aerospace sector to delve into data-driven approaches. One challenge in building a predictive model for aircraft engines is that the amount of in-service engine data for each engine varies. Furthermore, each engine experiences various operating conditions as it flies worldwide. For instance, when an engine is newly introduced, it is unlikely to have enough data to build a predictive model. This opens opportunities to adopt Transfer Knowledge [1], which is an approach that utilizes predictive models trained on existing engine data for new engines (Fig. 1).

Recent literature demonstrates that advanced ML models, including Long Short-Term Memory (LSTM) networks, tree ensembles and interpretable Generalized Additive Model

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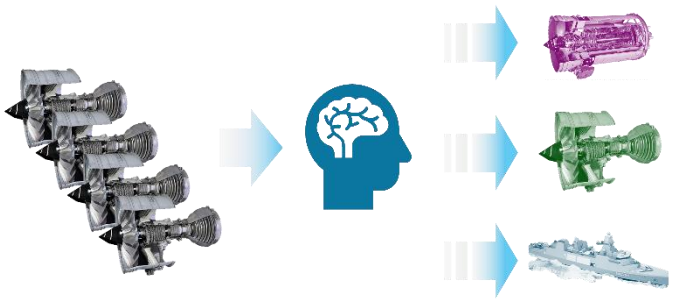


Figure 1 The concept of transfer knowledge across the engine types and applications

(GAM) [2], can predict Turbine Gas Temperature (TGT) or Exhaust Gas Temperature (EGT) with errors of only a few degrees Celsius for specific engines and datasets. Nevertheless, those publications exhibit several limitations that constrain their practical impact. First, most studies train and validate on limited sets of engines and operating conditions, so generalization and cross-fleet transferability remain largely untested [3]. Second, both physics-based and data-driven approaches are highly sensitive to sensor noise, bias, and missing data. There is a report shows that data quality improvements alone can influence TGT prediction RMSE by up to 50% [4]. Third, black-box ML models are difficult to interpret and certify in safety-critical settings, prompting interest in more transparent or physics-informed formulations [5]. Finally, there are very limited studies explicitly connect predicted health indices to maintenance decisions, Remaining Useful Life (RUL) estimation, and economic optimization, verifying that the link between algorithmic performance and operational value is still being established.

Cross-engine knowledge transfer in this paper was motivated by the concept of transfer learning. Transfer learning is an ML approach where a model trained on one task or source dataset is used for a new, related task or other target data. The possible benefits include, (1) efficiency, as it saves time for model training, (2) accessibility, as it adapts existing models to user requirement at a fraction of the cost, and (3) improved performance, as it often provides greater robustness. Recent large language models like Generative Pre-trained Transformers (GPT) are examples of transfer learning. In the aerospace industry, NASA C-MAPSS dataset [6, 7] is the predominant data source for ML-based predictive models for transfer learning [8–27]. Related research focuses on developing new ML algorithms or adopting different transfer learning methods, such as domain adaptation, and adversarial training. It should be noted that NASA C-MAPSS dataset only represents a single type of engine, and as a result, transfer learning has been demonstrated on the same engine type.

This paper proves the concept of transfer knowledge by evaluating the performance of the generalised predictive health model across the fleets of three types of turbofan engines. To support this validation, Engine Health Monitoring (EHM) datasets sourced from three turbofan family engines were utilized (Table 1). EHM data encompasses operational information from in-service engines, covering up to 20 years of operation for several hundred engines. The models were trained on each family engine separately and tested on each other, evaluating the prediction accuracies on TGT, which is a key health index for turbofan engines as it has a direct connection to performance and life degradation.

2. Methodology

2.1 Engine Health Monitoring Data

Table 1 EHM data summary for Turbofan family engines fleets

	Turbofan family engine A	Turbofan family engine B	Turbofan family engine C
Number of engines	Several hundred	Several hundred	Several hundred
Number of parameters	< 190	< 190	< 120
Flight segments	Taxi-out, Take-off, Climb, Cruise, Taxi-in		

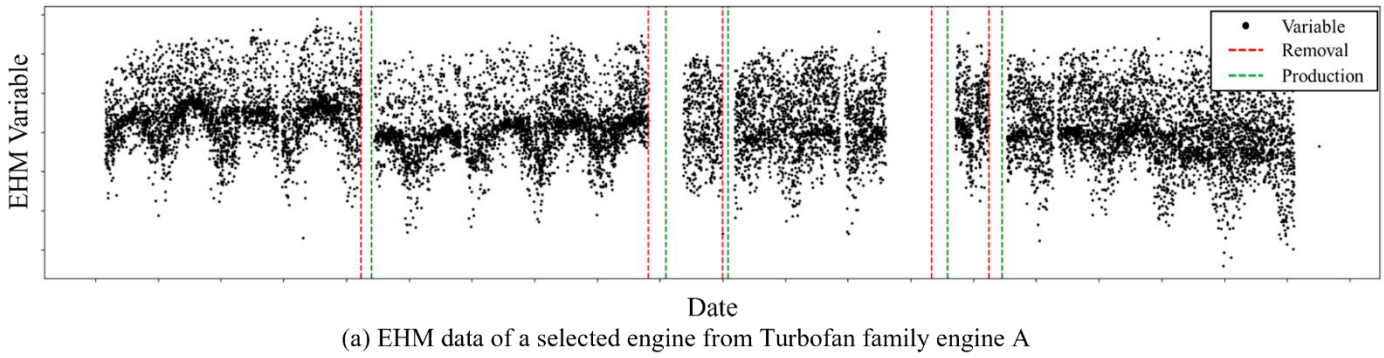
The EHM system on aircraft engine records installed sensor data at specific flight segments, including taxi-out, take-off, climb, cruise, and taxi-in. This is so-called ‘snapshot data’ which is discretely sampled data, collected over the service period of an engine. This data is critical to understand engine characteristics over its life cycle, including long-term performance degradation. The data includes, but not limited to, pressure, temperature, flow rate, vibration parameters, etc. In this paper, EHM data from several hundred engines across three turbofan family engines were utilized (Table 1).

Figure 2 illustrates a representative EHM variable over the period of service. The red and green vertical dotted lines show when an engine was removed from aircraft for major overhaul, and returned to service, respectively. It is important to note that an engine can be installed on a different aircraft after major overhaul. This implies that engine degradation patterns or characteristics are likely to change after major overhaul as it may operate under different routes and environments. The data may show gradual change in its baseline trend over continued engine operation. For example, seasonal behavior and abrupt changes of data pattern can be observed.

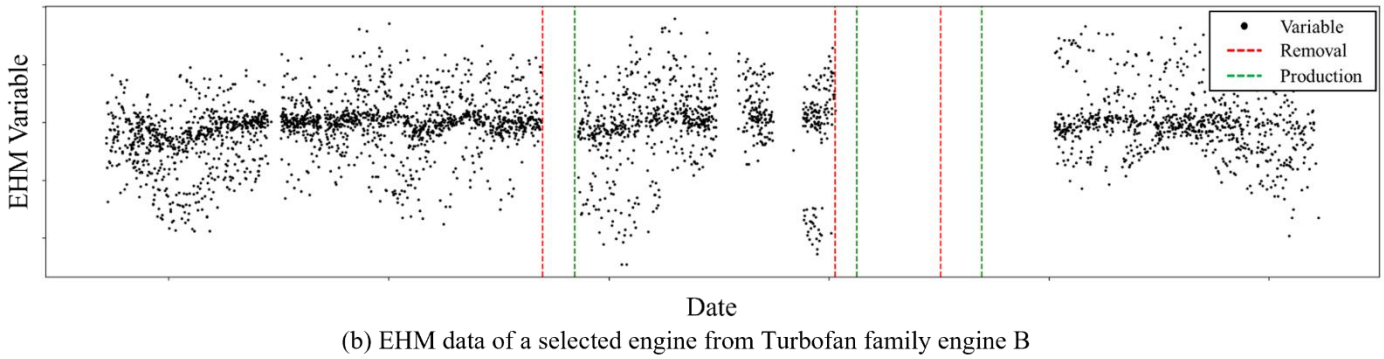
2.2 Detrending

Employing detrending can help models to focus on learning the underlying patterns separating the overall trend. In this paper, a polynomial fitting was applied in the pre-processing stage to subtract the trends in both train and test datasets. In the post-processing stage, the trend was added back to the prediction result. A separate sensitivity study showed that applying detrending improved the prediction accuracy by up to 85%, as measured by mean Root-Mean-Squared Error (RMSE).

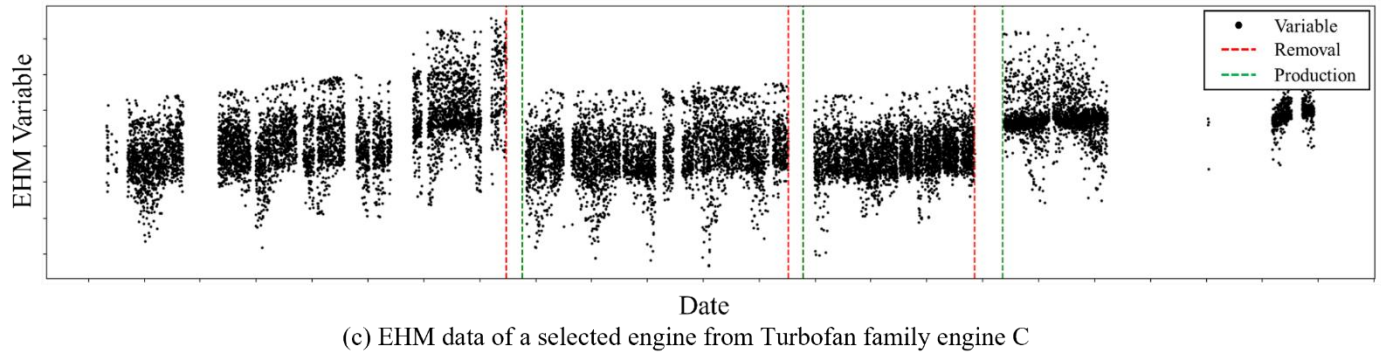
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(a) EHM data of a selected engine from Turbofan family engine A



(b) EHM data of a selected engine from Turbofan family engine B



(c) EHM data of a selected engine from Turbofan family engine C

Figure 2 Sampled EHM data and major overhaul records for selected engines across its life cycles

2.3 Long Short-Term Memory

Hochreiter and Schmidhuber [28] originally proposed Long short-term memory (LSTM) architecture to overcome the long-term dependency limitations in recurrent neural networks (RNNs) arising from a vanishing gradient problem [29]. The cell state (c_{t-1}, c_t) plays a key role as it improves a long-term dependency in recurrent units (Fig. 4). Since then, it has been widely used in machine learning approaches for predicting the remaining useful life of turbofan engines [8, 10, 12, 13, 15, 16, 20–22, 30–44]. Furthermore, it also showed robustness among available algorithms evaluated by authors' previous research [45]. Hence, LSTM was implemented to build ML-based predictive health model for the present study.

$$i_t = \sigma(W_{ii}x_t + b_{ii} + W_{hi}h_{t-1} + b_{hi}) \quad (1a)$$
$$f_t = \sigma\big(W_{if}x_t + b_{if} + W_{hf}h_{t-1} + b_{hf}\big) \quad (1b)$$
$$o_t = \sigma(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \quad (1c)$$
$$g_t = \tan\mathrm{h}\big(W_{ig}x_t + b_{ig} + W_{hg}h_{t-1} + b_{hg}\big) \quad (1d)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad (1e)$$
$$h_t = o_t \odot \tanh(c_t) \quad (1f)$$

2.4 Specific and Generalised Engine Health Models for Transfer Knowledge

A specific engine health model can be built by utilizing EHM data of a single or a fleet of engines. For transfer knowledge, it is necessary to have the same parameters in a train and test dataset, so that a model can be transferable. Table 2 shows the list of selected parameters for commonality and availability. The grid search method was used for the hyperparameter optimization of LSTM-based model. The hyperparameter sets were selected after a separate sensitivity study conducted prior to the model training. Hyperparameter sets were constructed from combinations of a single layer with 10 units, epoch counts of 100, 200, 500, and 1000, and batch sizes of 32, 64, and 128.

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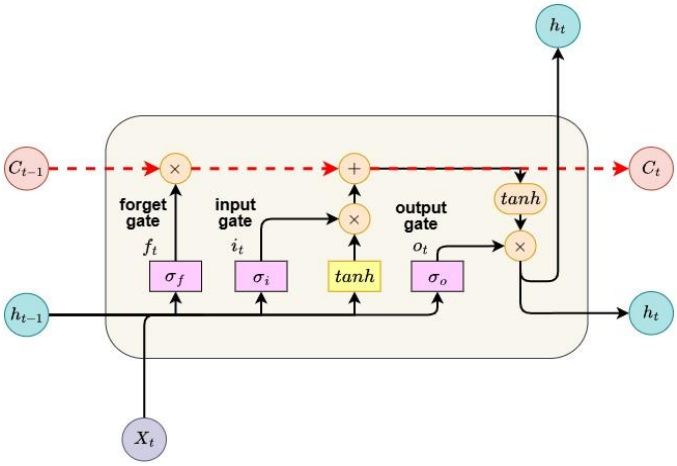


Figure 3 LSTM cell

All trials used a 0.001 learning rate, the Adam optimizer, and Mean Squared Error (MSE) loss function. Three specific engine health models, Engine A, B, C Model, were trained on the EHM datasets of three turbofan family engines, respectively while All Engine Model utilized all three train datasets, representing generalised engine health model.

To examine the potential of transfer learning (Figure 4) utilizing the specific and generalised engine health models, an analysis matrix was introduced as shown in Table 3. For instance, Engine A Model which was trained on the EHM data of Engine A made a prediction of TGT for Engines B, C (for transfer knowledge) and itself (baseline). This cross-validates the baseline and transfer knowledge cases of each Specific Engine Health Model.

Table 2 Selected EHM parameters

EHM parameter	Description	Unit
N1	Low pressure spool speed	%
N2	Intermediate pressure spool speed	%
N3	High pressure spool speed	%
P25	Intermediate pressure compressor delivery pressure	psi
P3	High pressure compressor delivery pressure	psi
T3	High pressure compressor delivery temperature	°C
TGT	Turbine gas temperature	°C

A generalised engine health model [45] will be suitable for applying transfer learning across the fleet of three family engines. Figure 5 shows the process to build the model applied for the present research. For this purpose, EHM data is segmented by major overhaul intervals and only the first segments of a family engine were collected resulting in three large training datasets. Each dataset was used for training the specific engine health model of each turbofan family engine. For generalised engine health model, All Engine Model was built utilizing all the three large training datasets.

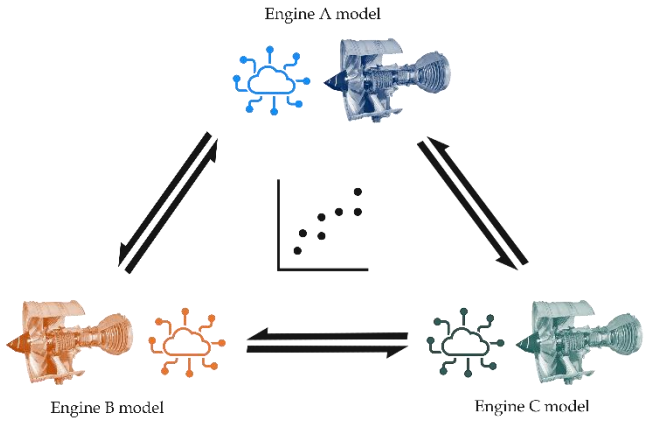


Figure 4 Overview of transfer knowledge among different turbofan family engines

Table 3 Transfer Knowledge Analysis Matrix for Specific and Generalised Models (Transfer knowledge cases are shaded in grey color)

Model\Target	Engine A	Engine B	Engine C
Specific Engine A Model	Model A on Engine A	Model A on Engine B	Model A on Engine C
Specific Engine B Model	Model B on Engine A	Model B on Engine B	Model B on Engine C
Specific Engine C Model	Model C on Engine A	Model C on Engine B	Model C on Engine C
Generalised Engine Model	All Engine Model on Engine A	All Engine Model on Engine B	All Engine Model on Engine C

3. RESULTS AND DISCUSSION

The accuracies of the predictive heath models were evaluated using RMSE (Equation 2) computed between predicted value and the target EHM data for each test dataset

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} |\hat{y}_i - y_i|^2}{N}} \qquad (2)$$

In this formulation, $\hat{y}$ denotes the predicted values, y represents the target EHM data, and N is the count of valid (non-missing) data points. RMSE was computed for test dataset, and the resulting mean and median values were used to assess the accuracy and quality of the predictions. To further evaluate performance and robustness, the interquartile range (IQR) and the ratio of outlier were reported as shown in Table 4.

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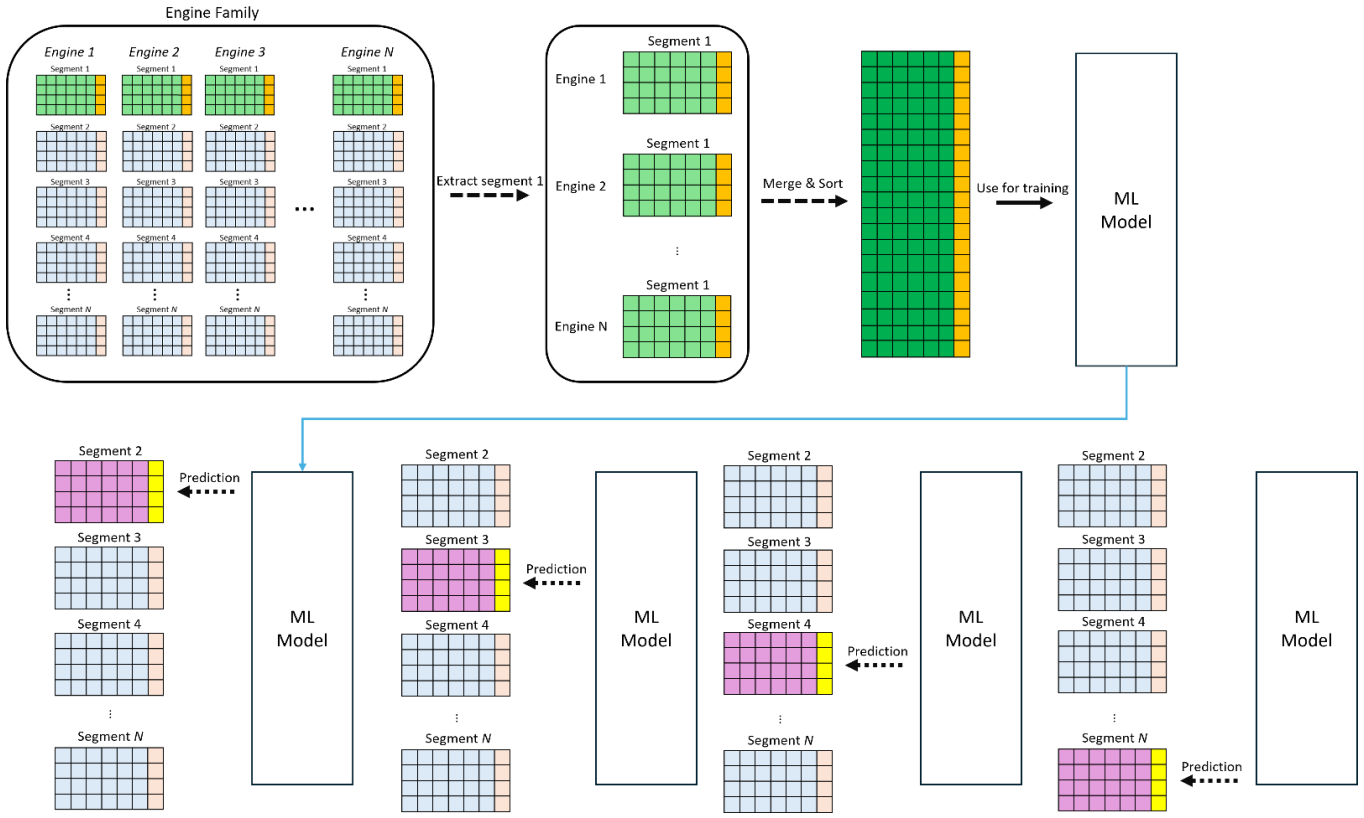


Figure 5 Generalised engine health model [45]

Table 4 Tabulated results of prediction quality (values in °C; Outlier in %; Transfer knowledge cases are shaded in grey color)

Model\Target	RMSE	Engine A	Engine B	Engine C
Specific Engine A Model	Mean	3.44	5.38	5.84
	Median	2.94	5.26	5.25
	Max.	30.48	14.42	16.19
	Min.	0.661	0.00	0.00
	IQR	0.94	2.08	2.30
	Outlier	9.47	2.74	3.70
Specific Engine B Model	Mean	6.85	3.38	6.49
	Median	5.19	3.28	5.93
	Max.	45.21	10.72	6.49
	Min.	0.00	2.11	0.00
	IQR	4.36	0.47	2.82
	Outlier	5.68	5.56	3.13
Specific Engine C Model	Mean	8.58	6.36	5.48
	Median	8.42	5.85	4.88
	Max.	41.41	23.54	23.92
	Min.	0.00	0.00	1.72
	IQR	6.39	1.66	1.64
	Outlier	1.84	7.96	8.19
Generalised Engine Model	Mean	8.14	7.42	7.69
	Median	6.85	6.72	7.26
	Max.	31.94	21.04	19.65
	Min.	0.00	0.00	0.00
	IQR	4.08	4.11	2.66
	Outlier	4.17	1.25	3.13

Figure 6 depicts box plots and violin plots of the prediction accuracy and quality. In the box plots, the vertical axis represents RMSE values in °C. The 'x' marker denotes the mean, while the horizontal line within each box indicates the median. The box height of the box corresponds to the IQR defined as the span between 25th and 75th centiles. Whiskers extend to the lowest and highest non-outlier points, with outliers displayed as hollow circles. The violin plots describe the distributions of the prediction results which provide useful insight on the results with the box plots.

Overall, specific engine health model showed the best prediction accuracy and quality when it was tested on the same family engine, i.e., the baseline cases as described in Table 3. However, it is worth highlighting that the transfer knowledge cases, also produced comparable predictions to each other. The mean RMSE values of these cases are within 5°C – 9°C, while the accuracies of the baseline cases are in the range of 3.44°C to 5.48°C. The IQR values of the baseline cases are the smallest, ranging from 0.5°C to 1.6°C and the transfer knowledge cases show IQR values between 1.6°C and 6.4°C. This implies that there is less variation in the prediction results with the baseline cases than the transfer knowledge ones.

Although the amount of the outliers seems to be increased for the baseline cases, the models did not perform worse than that of the transfer knowledge cases. This is because there are less variations in the prediction results than the transfer knowledge cases.

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Generalised Engine Model offers insight into how a unified model performs when trained on heterogeneous data. Its mean RMSE values (7.42°C to 8.14°C) fall within the upper range of the transfer knowledge cases, indicating that while it cannot match the precision of the engine-specific models in their normal cases, it still delivers reasonably consistent predictions across the three family engines. The IQR values (2.66°C to 4.11°C) are larger than those of the baseline cases but comparable to several transfer scenarios, reflecting the broader variability the model must accommodate. Despite this, the outlier ratios (1.25% to 4.17%) remain similar to or lower than those observed in the transfer knowledge cases, suggesting that the unified model does not produce disproportionately extreme errors. Therefore, Generalised Engine Model, a cross-engine framework can also achieve reliable and competitive performance, reinforcing the feasibility of deploying a unified predictive model when engine-specific optimization is not available or essential.

Overall, the results demonstrated the potential of transfer knowledge of the predictive health model among turbofan family engines. A possible use case may be a newly introduced engine that is unlikely to have the amount of data to build a predictive health model. While EHM system on the new engine will start collecting data as it operates, transfer knowledge from specific and generalised engine health models can provide a reliable predictive engine health model that is pre-trained on other available engine data.

Figure 7 shows the selected comparisons on TGT predictions. Shown are the actual readings from the EHM data (black dots) and the predicted TGT values (red circles). Normalization is applied on TGT at the beginning of the dataset (or service), as specified in Equations 3 and 4. The horizontal axis shows the date associated with flight history.

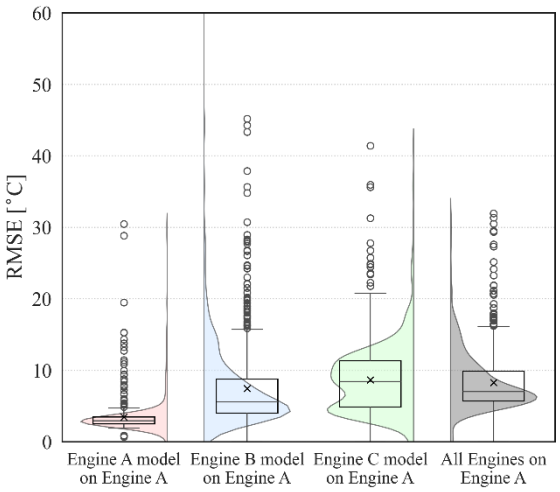
$$\mathrm{TGT}_{\mathrm{norm,\ EHM}} = \frac{\mathrm{TGT}_{\mathrm{EHM}}}{\mathrm{TGT}_{\mathrm{EHM,\ 1st\ date}}} \qquad (3)$$

$$\mathrm{TGT}_{\mathrm{norm,\ predicted}} = \frac{\mathrm{TGT}_{\mathrm{predicted}}}{\mathrm{TGT}_{\mathrm{EHM,\ 1st\ date}}} \qquad (4)$$

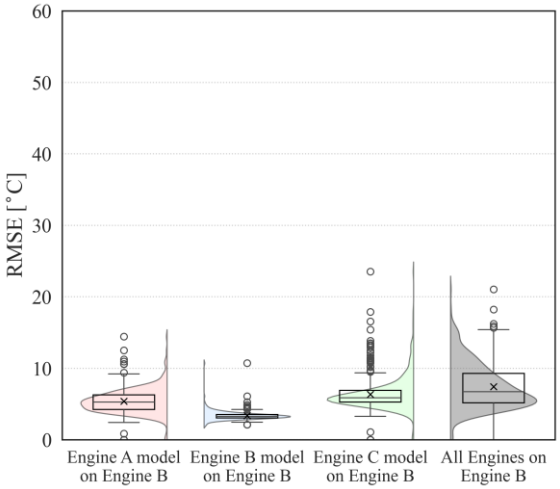
Figure 7a shows the prediction results of the selected segments with near-mean RMSE value from the Engine A Model tested on the dataset of Engine B, and Figure 7b represents an outlier segment. In general, the model predicted the trends of the target TGT values in both cases, while the accumulation of local deviations in values still influenced RMSE.

4. CONCLUSION

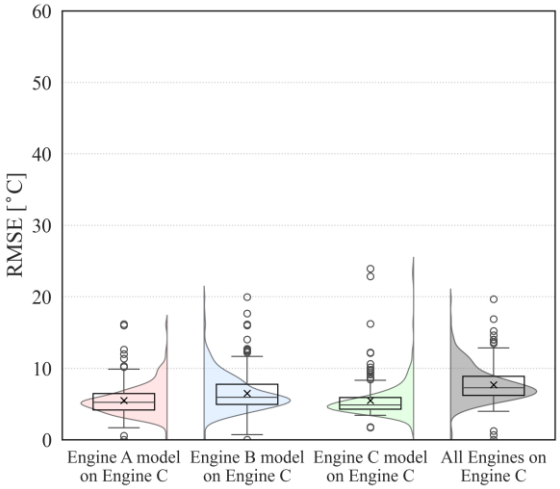
The concept of transfer knowledge in the ML-based predictive health model was evaluated using the EHM datasets of three Turbofan family engines. The prediction accuracies of specific and generalised engine heath models in mean RMSE values were in the range of 5°C to 8°C. Although the values are about 2 times higher than the baseline cases, this is justifiable for a generalisable predictive health model. The transfer knowledge cases showed wider variation in the prediction



(a) Models tested on Engine A



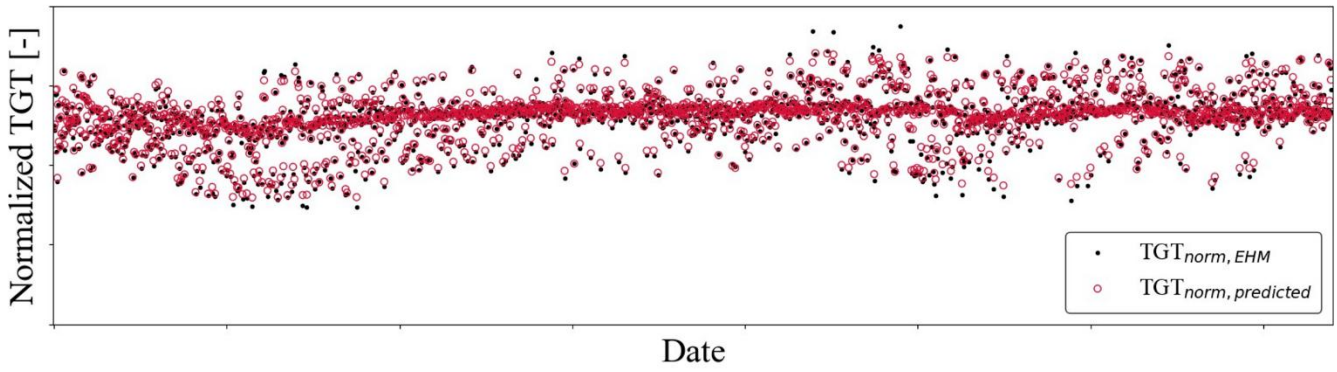
(b) Models tested on Engine B



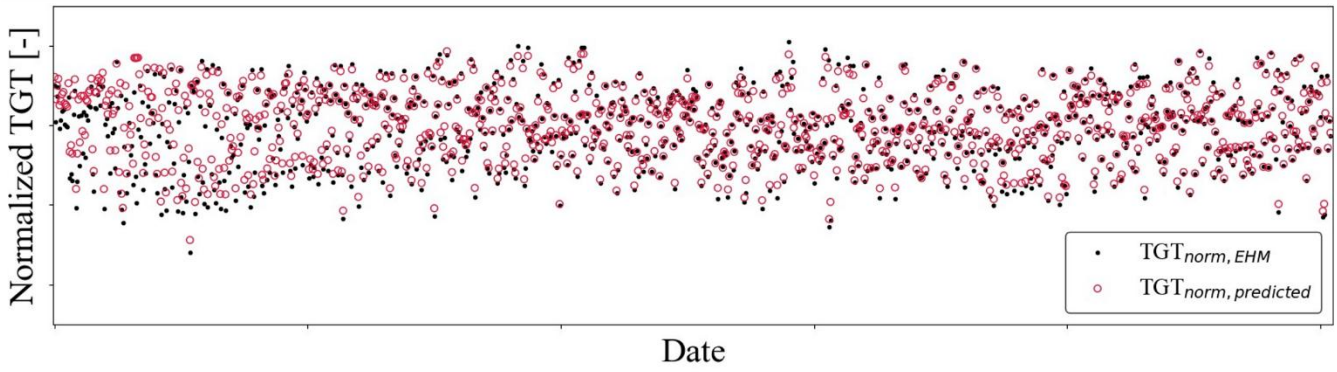
(c) Models tested on Engine C

Figure 6 Box and violin plots of the results from the baseline and transfer knowledge cases on each engine

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(a) Segment with near-mean RMSE value



(b) Segment with near-outlier RMSE value

Figure 7 Comparisons on TGT predictions

accuracy than the baseline case; however, it demonstrated the versatility of this approach. It is also worth highlighting that this approach would not require additional data to re-train the model for different applications. Therefore, the proposed generalised engine health model will benefit use cases beyond three Turbofan family engines, such as providing a pre-trained predictive health model for newly introduced engines which can forecast its safe and economic operation.

ACKNOWLEDGEMENTS

The authors would like to express sincere appreciation to Rolls-Royce for sharing EHM data and technical insights. The financial support was from Virginia Tech and Rolls-Royce.

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